

Zeqian Long

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RESEARCH STATEMENT

My research focuses on **generative AI**. I work on controllable visual synthesis, adversarial robustness, and structured representation learning for modeling world dynamics in precise, trustworthy, and grounded ways.

EDUCATION

May 2026	University of Illinois Urbana–Champaign	IL, United States
Aug 2022	B.S. in Mathematics and Computer Science (Dean's List)	GPA: 4.0 / 4.0
Coursework (* = graduate-level): Deep Generative Models*, Text Analysis with LLMs*, Graph Learning & Knowledge Mining*, Statistical Machine Learning*, Advanced Algorithms*, Theoretical Mathematics (Stochastic Process, Real Analysis, Abstract Algebra)		

PUBLICATIONS

S=IN SUBMISSION (* = EQUAL CONTRIBUTION)

- [S.1] **Follow-Your-Shape / EditAnyShape: Shape-Aware Image Editing via Trajectory-Guided Region Control**
Zeqian Long*, Mingzhe Zheng*, Kunyu Feng*, Xinhua Zhang, Hongyu Liu, Harry Yang, Linfeng Zhang, Qifeng Chen, Yue Ma
Under Review at ICLR 2026 (scores: 6, 6, 4, 6) [[arXiv](#) | [GitHub](#) | [OpenReview](#)]

RESEARCH EXPERIENCES

Shape-Aware Image Editing via Trajectory-Guided Region Control MAR'25 – SEP'25

EPIC Lab, Shanghai Jiao Tong University, supervised by *Yue Ma, Prof. Linfeng Zhang*

- Developed a training- and mask-free image editing framework built on FLUX.1-dev that dynamically localizes editable areas and enables precise and large-scale shape transformations.
- Demonstrated state-of-the-art performance on PIE-Bench and our constructed 290-case ReShapeBench, outperforming FLUX.1 Kontext by over 10% across key quality and fidelity metrics.
- **Github:** <https://github.com/mayuelala/FollowYourShape>; **Status:** First author, paper in submission.

Attention-Driven Adversarial Attack for Image-To-Video Models JUL'25 – PRESENT

UIUC, collaborated with *Haotian Xue, Ozgur Kara*

- Proposed a memory-efficient adversarial attack framework for the Image-to-Video model [Wan2.1-14B](#), inducing significant drops in semantic consistency and degradation in temporal coherence and visual fidelity.
- Designed a cross-attention-driven semantic loss that contrasts source-target CLIP and latent attention to extract semantic deviation signals; integrated multi-resolution VAE encoder feature loss that amplifies reconstruction errors across scales; applied PGD to optimize adversarial perturbations on input images.
- **Github:** <https://github.com/Zeqian-Long/Wan-I2V-Attack>; **Status:** Planned submission to **ICML 2026**.

Ontology Grounded Scientific Graph Construction for LLM Retrieval and Reasoning AUG'25 – PRESENT

ULab, UIUC, supervised by *Prof. Jiaxuan You, Prof. Jiawei Han*

- Developed a hierarchical scientific ontology that organizes research papers into interpretable semantic dimensions (task, method, dataset, architecture, evaluation, contribution).
- Constructed a fine-grained concept space by designing a corpus-driven induction pipeline that extracts, normalizes, and clusters meaningful semantic fragments, enabling accurate and interpretable paper-to-concept mapping.
- Leveraged the ontology to derive paper relations that reveal deeper connections beyond citations, directly improving LLM fine-grained retrieval, multi-hop scientific QA, and cross-paper comparison analysis.

Neuro-Symbolic Knowledge Graph for Aligning LLM Worldviews AUG'25 – PRESENT

UIUC, supervised by *Prof. Jiawei Han*

- > Constructed a neuro-symbolic framework that constructs a Global Values Knowledge Graph by integrating quantitative survey data (WVS, EVS), legal texts, and cultural corpora from 50+ countries to support verifiable cross-cultural and cross-lingual alignment of LLM worldviews.
- > Developed a KG-grounded RAG reasoning pipeline that leverages the Knowledge Graph to improve factual grounding, cross-cultural consistency, and temporal awareness in LLM reasoning over value-sensitive queries.

PROJECTS

Predicting Crop Yield based on Sentinel-2 Satellite Data MAR'24 – JUN'24

Independent Project

- > Preprocessed Sentinel-2 satellite imagery by aligning it with European crop type maps, sampling consistent spatial blocks, and normalizing spectral features across growth periods.
- > Applied modern dimensionality reduction to convert preprocessed images into histograms; trained a CNN and employed Deep Gaussian Process to extract key spectral-temporal patterns for preliminary crop yield estimation across 500 regional samples.

SmartRide – Smart Cycling App for Smarter Riders JAN'25 – MAY'25

UIUC, course project, full-stack development

- > Designed a smart cycling platform integrating route planning, real-time tracking, and personalized user profiling.
- > Implemented a route planning module with dynamic map generation, distance optimization, and turn-by-turn navigation; built a route persistence system enabling save, edit, and reload of user-defined paths.
- > Github: <https://github.com/CS222-UIUC/team-07-project-smartride>

WORKING EXPERIENCES

Dec 2024 | **UIUC Teaching Assistanship | CS 446: Machine Learning** **IL, United States**

Aug 2024 | *Course Assistant — Professor: [Liangyan Gui](#)*

- > Assist students with machine learning concepts; proofread and grade the homework and exams; manage office hours; support course operations to enhance learning outcomes.

Aug 2024 | **Beijing Academy of Artificial Intelligence | Vision and Multimodal Research Center** **Beijing, CN**

Jun 2024 | *Research Intern — Advisor: [Xinlong Wang](#)*

- > Reproduced 10+ diffusion and autoregressive image generation papers, examining architectural design choices and organizing experimental workflows.
- > Contributed to multimodal visual generation research within the [Emu](#) framework by evaluating baseline models, assessing cross-modal generation quality, and maintaining experiment documentation.

SKILLS

Programming	Python (PyTorch, TensorFlow), C/C++, Java, Javascript, Bash
AI/ML Frameworks	PyTorch Lightning, HuggingFace Transformers, Diffusers, OpenCV, scikit-learn
Systems/Tools	Linux, Git, Docker, CUDA Toolkit